Run report

2026-10-30T203017Z-1

Configuration

Parameter	Value	Parameter	Value
batch_size	1,024	n_samples	500,000
dim	4	dataset	gaussian
seed	117,039,692	data_seed	42
hidden_units	64	n_layers	2
n_epochs	100	n_disc_steps	5
lr_g	3.000×10^{-5}	lr_d	1.000×10^{-4}
lambda_dispersion	0.01500	mmd_subsample	16,384
mmd_test	2.984×10^{-3}	best_epoch	84
compile_cache_warm	True		
gaussian_params	dim=4, mu_gen=[1, 0, -0.5, 0.5], mu_true=[0.8, 0.1, -0.6, 0.7], cov_gen=[[1, 0.07, -0.22, 0.24], [0.07, 0.49, 0, 0.056], [-0.22, 0, 1.21, 0.616], [0.24, 0.056, 0.616, 0.64]], cov_true=[[0.64, 0, -0.24, 0.192], [0, 0.36, 0.12, 0], [-0.24, 0.12, 1, 0.3], [0.192, 0, 0.3, 0.36]], cov_detector=[[0.16, 0, 0, 0], [0, 0.25, 0, 0], [0, 0, 0.36, 0], [0, 0, 0, 0.09]]		
mmd_sigmas_detector	$1.590 \times (\frac{1}{2},\ \frac{1}{\sqrt{2}},\ 1,\ \sqrt{2},\ 2)$		
mmd_sigmas_particle	$1.336 \times (\frac{1}{2},\ \frac{1}{\sqrt{2}},\ 1,\ \sqrt{2},\ 2)$		

Timing

Phase	Seconds	Share (%)	Detail
Deconvolve			
data	0.205	0.6	gaussian cache hit – train
train	24.765	77.4	train
transfer	6.496		train
compile	2.923		train
epochs	6.528		train
select	8.179		train
particle_mmd	0.519	1.6	train
save	0.108	0.3	train
evaluate	6.385	20.0	train
total	31.981	100.0	
IBU baseline			
parse_config	0.001	0.0	ibu
data	0.173	2.7	gaussian cache hit – ibu
ibu	0.308	4.8	niter=10 – ibu
evaluate	5.874	92.4	4 variables – ibu
total	6.357	100.0	
OmniFold baseline			
parse_config	0.001	0.0	omnifold
data	0.176	0.0	gaussian cache hit – omnifold
omnifold	417.364	98.7	niter=3, epochs=50 – omnifold
init	4.966		DataLoaders and the two MLPs – omnifold
unfold	376.719		MultiFold.Unfold – omnifold
iter1_step1	78.004		detector-level reweighting – omnifold
iter2_step1	70.757		detector-level reweighting – omnifold
iter3_step1	58.029		detector-level reweighting – omnifold
iter1_step2	64.779		particle-level reweighting – omnifold
iter2_step2	58.446		particle-level reweighting – omnifold
iter3_step2	46.672		particle-level reweighting – omnifold
reweight	1.580		evaluating weights on z_target – omnifold
evaluate	5.378	1.3	4 variables – omnifold
total	422.920	100.0	
Common			
parse_config	0.001	0.0	plot
data	0.129	0.6	gaussian cache hit – plot
load	1.330	6.4	plot
plots	19.190	92.9	plot
total	20.650	100.0	

Detector-level metrics

Distance to Data. Improvements are relative to Simulation.

Wasserstein Distance [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
dim_0	231.7	—	—	**16.76**	**92.8**	29.26	87.4
dim_1	107.6	—	—	22.98	78.6	**17.79**	**83.5**
dim_2	116.2	—	—	**65.35**	**43.8**	70.44	39.4
dim_3	218.7	—	—	**23.87**	**89.1**	37.67	82.8

JS divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
dim_0	13.83	—	—	**0.3933**	**97.2**	0.6000	95.7
dim_1	4.552	—	—	0.5221	88.5	**0.4281**	**90.6**
dim_2	2.293	—	—	0.8505	62.9	**0.8029**	**65.0**
dim_3	20.87	—	—	**0.3888**	**98.1**	0.7470	96.4

VLC divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
dim_0	52.60	—	—	**1.538**	**97.1**	2.335	95.6
dim_1	17.80	—	—	2.054	88.5	**1.676**	**90.6**
dim_2	9.023	—	—	3.347	62.9	**3.169**	**64.9**
dim_3	78.36	—	—	**1.536**	**98.0**	2.952	96.2

Sliced Wasserstein, joint [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
All	227.5	—	—	**33.21**	**85.4**	42.67	81.2

Mean 1D distance over random directions, in Data standard deviations. IBU: product of its per-observable weights.

Particle-level metrics

Distance to Truth. Improvements are relative to Generation.

Wasserstein [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
dim_0	240.2	—	—	**17.42**	**92.7**	30.69	87.2
dim_1	117.8	—	—	29.68	74.8	**21.33**	**81.9**
dim_2	121.0	—	—	**63.98**	**47.1**	70.12	42.0
dim_3	226.5	—	—	**19.74**	**91.3**	36.38	83.9

JS divergence [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
dim_0	18.41	—	—	**0.3369**	**98.2**	0.6703	96.4
dim_1	9.436	—	—	1.059	88.8	**0.8444**	**91.1**
dim_2	3.592	—	—	1.145	68.1	**1.092**	**69.6**
dim_3	27.10	—	—	**0.3169**	**98.8**	0.8987	96.7

VLC divergence $[\times 10^3]$

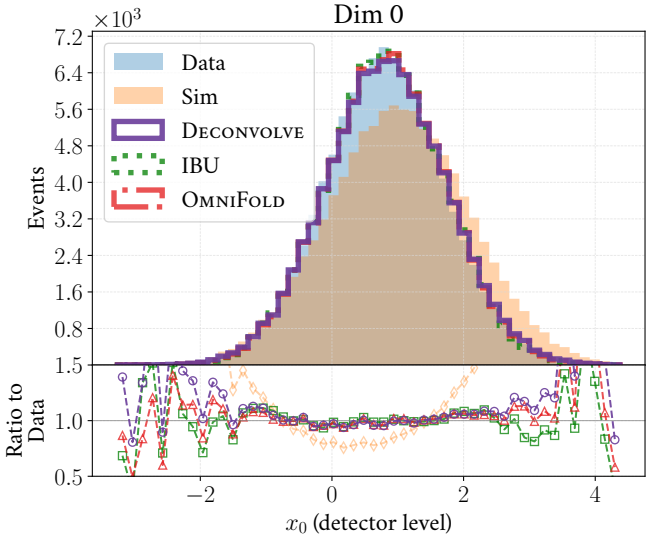
Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
dim_0	69.53	—	—	**1.322**	**98.1**	2.627	96.2
dim_1	36.25	—	—	4.154	88.5	**3.279**	**91.0**
dim_2	14.07	—	—	4.506	68.0	**4.324**	**69.3**
dim_3	100.6	—	—	**1.247**	**98.8**	3.537	96.5

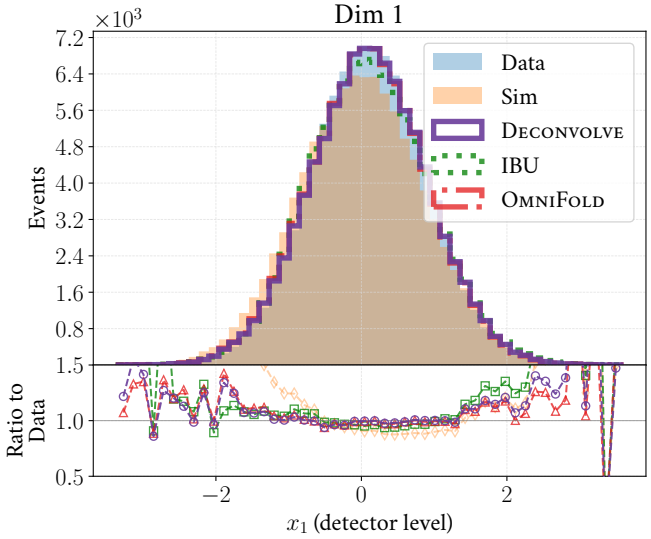
Sliced Wasserstein, joint $[\times 10^3]$

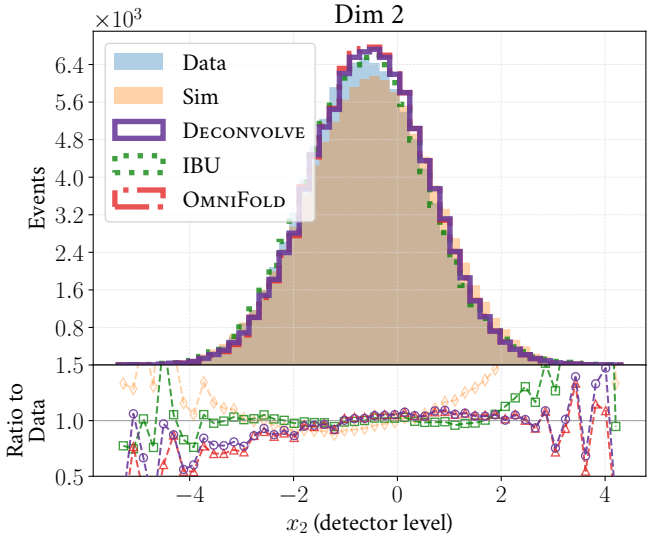
Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
All	272.6	—	—	**38.48**	**85.9**	50.59	81.4

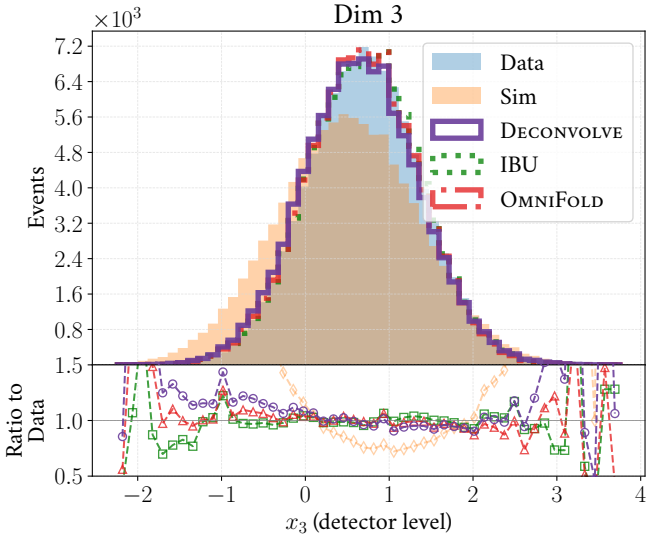
Mean 1D distance over random directions, in Truth standard deviations. IBU: product of its per-observable weights.

Detector Level

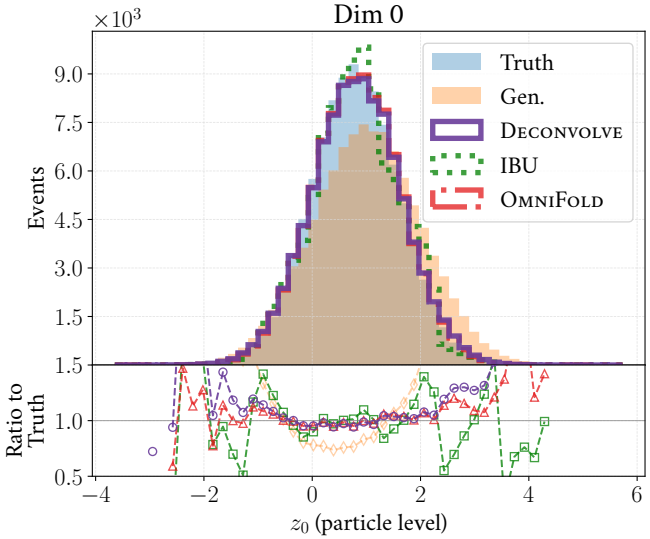


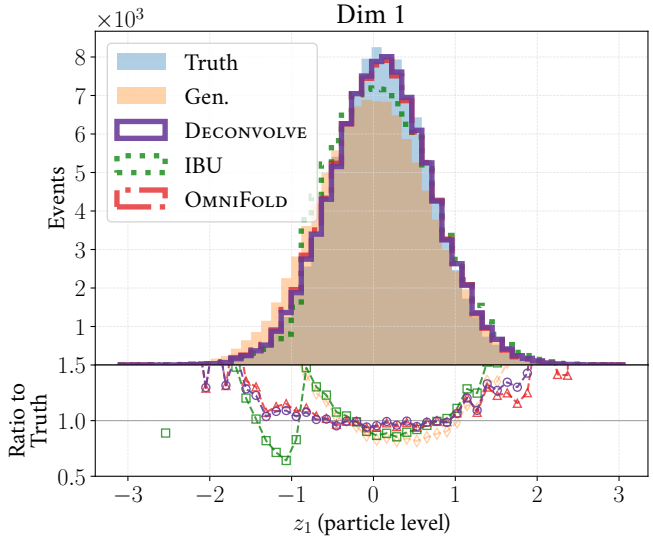


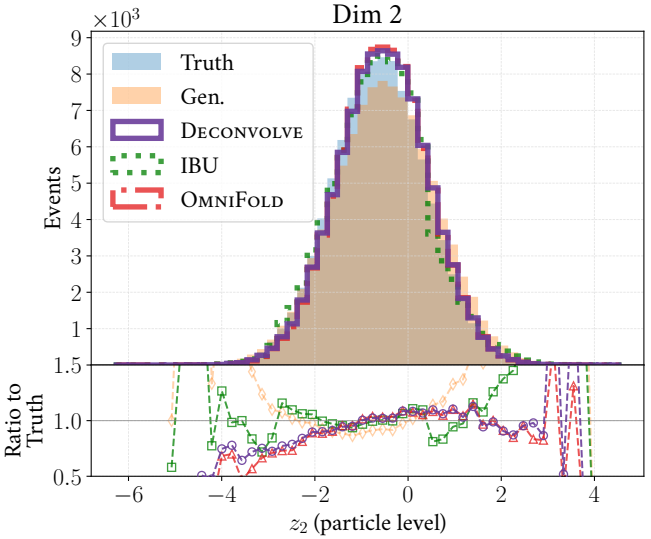


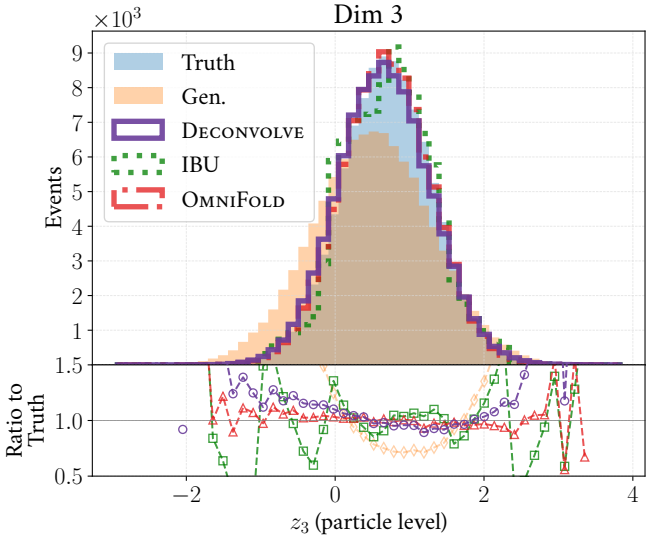


Particle Level



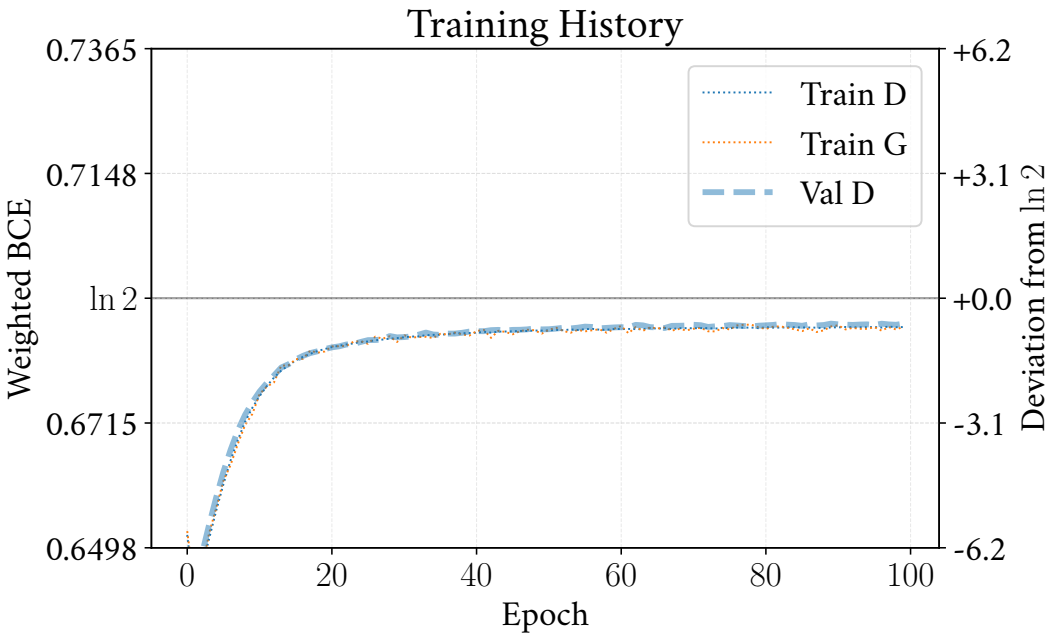






Training diagnostics

Training losses



Selection criteria

